

HW4 Findings - Mini GPT from Scratch

Dataset: shakespeare.txt

Vocabulary size: 49

Sequence length: 128

Hidden dimension: 128

Number of heads: 4

Number of layers: 2

Total parameters: 425,777

Epochs: 5

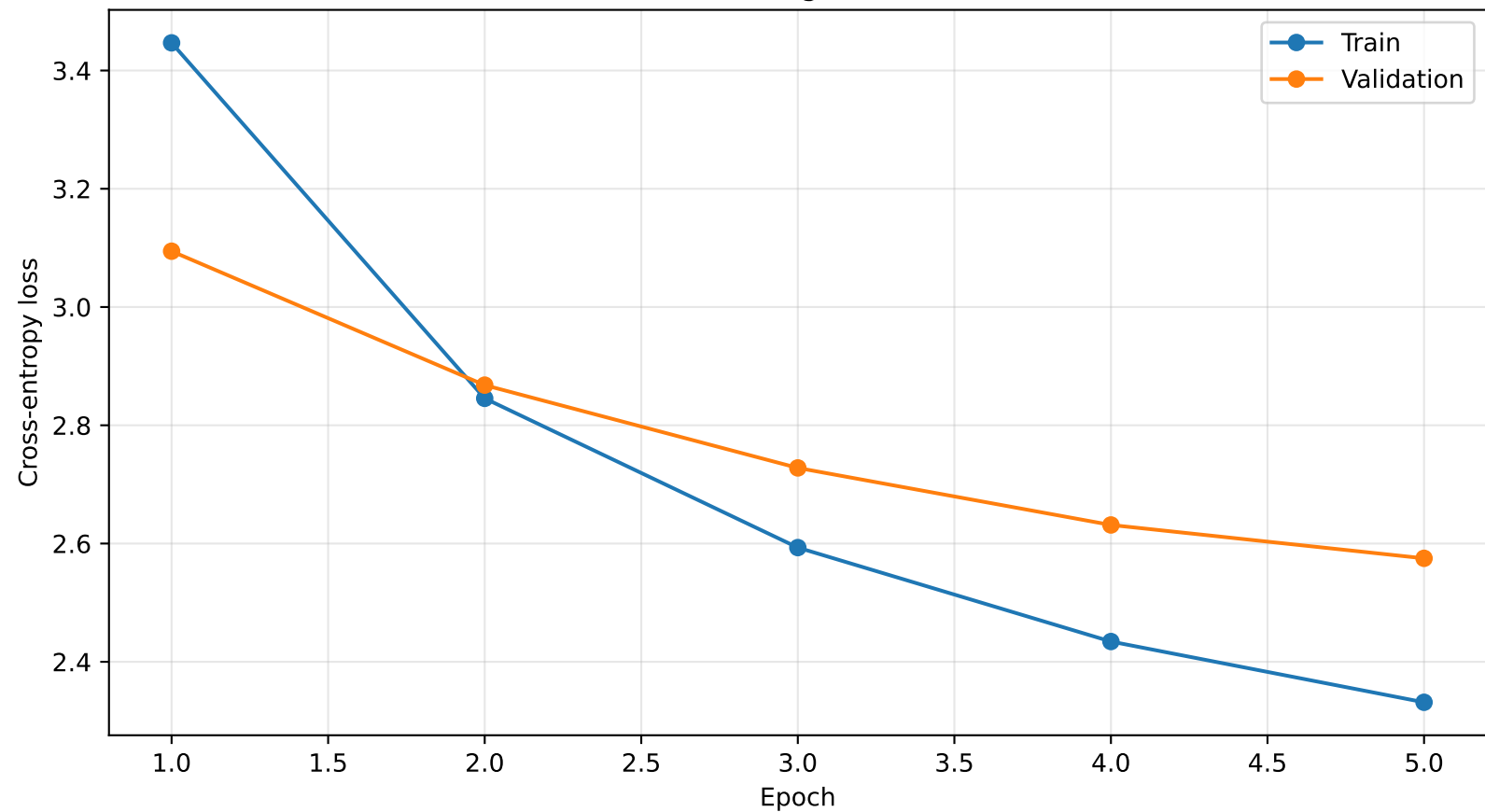
Learning rate: 0.0003

Batch size: 64

Final train loss: 2.3315

Final validation loss: 2.5750

Training Loss



Decoding Analysis

Greedy decoding chooses the most likely next character at every step. It is usually coherent but can repeat common patterns.

Temperature sampling changes randomness. Temperature 0.5 is conservative, 1.0 is balanced, and 1.5 is more diverse but less reliable.

Top-k sampling samples only from the k most likely next characters. k=10 is more focused, while k=50 is more diverse.

For this small character model, greedy or temperature 0.5 usually gives the most coherent output. Temperature 1.5 and top-k 50 usually produce the most diverse output.